

ADAM v30: ESP32-CAM Architecture Guide

Because the ESP32-CAM dedicates almost all of its pins to the camera module, squeezing in 4 touch sensors, an I2C LiDAR, and a servo requires precise pin management.

1. Wiring & Pinout Map

Component	ESP32-CAM Pin	Important Notes
VL53L0X (SDA)	GPIO 14	Connect to SDA. Safe to use.
VL53L0X (SCL)	GPIO 15	Connect to SCL. Safe to use.
Tilt Servo (Signal)	GPIO 13	Connect to orange/yellow PWM wire.
Touch Sensor 1	GPIO 2	CRITICAL: TTP223 must output LOW when untouched. Do not touch during boot.
Touch Sensor 2	GPIO 12	CRITICAL: Will prevent boot if pulled HIGH. TTP223 defaults to LOW, so it is safe as long as untouched during boot.
Touch Sensor 3	GPIO 3 (U0RXD)	Unplug this sensor while uploading code to the ESP32 via your programmer.
Touch Sensor 4	GPIO 1 (U0TXD)	Unplug this sensor while uploading code to the ESP32 via your programmer.
Power (All)	5V / GND	DO NOT power the MG90S servo from the ESP32-CAM's pins. Use a

		dedicated 5V rail as noted in the v2.2 component spec, or your ESP32 will constantly crash due to brownouts.
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2. Arduino IDE Setup Requirements

Before clicking upload on the C++ code, ensure your Arduino IDE is set up properly for the AI Thinker module:

1. **Board:** "AI Thinker ESP32-CAM"
2. **PSRAM:** MUST be set to "Enabled" (Required for the camera's double buffering stream).
3. **Libraries Required:** * ESP32Servo by Kevin Harrington
 - o Adafruit_VL53LOX by Adafruit

3. How the "Trick" Works

Because we had to use the TX (1) and RX (3) pins for your 3rd and 4th touch sensors, you cannot use the Arduino Serial Monitor permanently.

When the ESP32-CAM boots, it connects to Wi-Fi and prints its IP address to the Serial Monitor. **You have 4 seconds to write down this IP address.** It then immediately kills the Serial connection and converts those two pins into standard digital inputs so your Touch Sensors can work.

4. Testing Your Endpoints

Once running, verify the ESP32 is working from a browser on the same Wi-Fi network:

- **Camera:** <http://192.168.1.13/stream> (You should see live video)
- **Sensors:** <http://192.168.1.13/sensors> (You should see {"touch":[0,0,0,0], "lidar_mm":150})
- **Servo:** <http://192.168.1.13/tilt?angle=100> (The neck should move)